



Anticancer potential of exosome-like nanoparticles isolated from *Acorus calamus* in breast cancer

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Abstract

Plant-derived exosome-like nanoparticles represent a novel class of plant-based therapeutics with potential anti-cancer applications. The present study aims to isolate, characterize, and evaluate the *Acorus calamus*-derived exosome-like nanoparticles (ACENPs) for their cytotoxic and apoptotic effects on breast cancer cells. Nanoparticle tracking analysis (NTA) and transmission electron microscopy (TEM) revealed that ACENPs exhibited a mean hydrodynamic diameter of 122.4 ± 5.0 nm and a particle concentration of 1.58×10^{11} particles/mL. Cellular uptake studies confirmed the efficient internalization of ACENPs in MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells. Cytotoxicity assay demonstrated a significant reduction in cell viability by 17.6%, 25%, and 35.8% in MCF-7, MDA-MB-231, and MDA-MB-453 breast cancer cells, respectively. Apoptosis induction was validated through AO/EB staining, DAPI nuclear fragmentation assays, annexin V-FITC/PI staining, and Western blot analysis of apoptosis-related proteins. Treatment with ACENPs resulted in an increased Bax/Bcl-2 ratio, indicating apoptotic activation. Metabolomic profiling identified bioactive compounds such as arecoline, trigonelline, asarone, and gingerol, known for their anti-cancer properties. Our study findings highlight that ACENPs could be utilized as a promising therapeutic approach for breast cancer treatment. We propose to conduct future research focusing on in vivo validation and optimizing large-scale production for clinical translation.

Keywords Breast cancer · Exosome · Apoptosis · Therapeutics · Nanoparticles

Introduction

Breast cancer is a heterogeneous disease classified based on its molecular diversity and is the most common cause of cancer-related death among women worldwide. Molecular classification of breast cancer is based on the expression levels of estrogen receptor (ER), progesterone receptor (PR), human epidermal growth factor receptor 2 (HER2),

and the proliferation marker Ki67. Based on these molecular markers, breast cancer is classified into Luminal A, Luminal B, HER2-enriched, and triple-negative (TNBC) subtypes (Johnson et al. 2021). HER2-positive and TNBC are generally associated with higher histological grades and increased tumor size than luminal A and B breast cancers (Al-thoubaty 2020). The treatment regimen for breast cancer depends on its molecular subtype. Strategies used for treating luminal breast cancer include a combination of aromatase inhibitors (anastrozole, letrozole, tamoxifen, etc.) and chemotherapy (Munzone and Colleoni 2018). Comparatively, HER2-positive breast cancer is treated with chemotherapy along with targeted therapy (trastuzumab, pertuzumab, T-DM1, etc.) (Pondé et al. 2018). Due to the absence of ER, PR, and HER2 receptors, triple-negative breast cancer (TNBC) is difficult to treat and lacks effective treatment strategies (Chang-Qing et al. 2020).

Despite the availability of different treatment strategies for breast cancer, increasing resistance to existing treatments and the side effects of chemotherapy necessitate

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new treatment options (Bai et al. 2018; Hanker et al. 2020; Masoud & Pagès, 2017; Nurgali et al. 2018).

Exosomes are extracellular vesicles of nanometer size (30–150 nm) secreted by eukaryotic cells and present in nearly all body fluids. They play a crucial role in intercellular communication by mediating the transfer of molecular cargo. In a clinical trial, exosomes isolated from dendritic cells (Dex) were observed to enhance natural killer cell function in patients with non-small cell lung carcinoma (Besse et al. 2016). Several reports suggest that exosomes can be an effective drug delivery system for cancer treatment (Kalimuthu et al. 2018; Kim et al. 2016; Schindler et al. 2019). Exosome-loaded doxorubicin exhibits higher therapeutic potency than the free form of doxorubicin (Schindler et al. 2019). In another study, paclitaxel loaded into macrophage-derived exosomes exhibited 50-fold enhanced cytotoxicity in (MDCKMDR1) multi-drug resistance cells (Kim et al. 2016). A similar study reported that exosomes isolated from mesenchymal stem cells loaded with paclitaxel showed significant anti-proliferative activity in MDA-MB-231 breast cancer cells. In addition, tumor growth was inhibited in mice models after treatment with paclitaxel-loaded exosomes (Kalimuthu et al. 2018). Although drug-loaded exosomes have demonstrated excellent anticancer potential, their clinical application remains limited due to their low yield. In addition, contaminants such as endotoxins and mycoplasma present in fetal bovine serum further restrict their use in clinical settings (Urzi et al. 2022). To overcome these limitations, an alternative source of extracellular vesicles is needed.

Recently, plant-derived exosomes have become an excellent alternative to mammalian cell culture-derived exosomes. Plant-derived exosome-like nanoparticles (PDENPs) offer several advantages over conventional cancer therapeutics. PDENPs are biocompatible, biodegradable, and less toxic than chemotherapeutics because they originate from natural sources. Due to their low immunogenicity, PDENPs could be employed as therapeutic vehicles for drug delivery (Nemati et al. 2022).

Several studies reported the potential of plant-derived exosomes like nanoparticles as an anticancer agent. In a study, exosomes isolated from *Citrus limon* L. inhibited the growth of lung cancer cells (A549), human colorectal adenocarcinoma cells (SW480), and human chronic myeloid leukemia cells (LAMA84) in a dose and time-dependent manner. Mechanistic insights revealed that *C. limon*-derived nanovesicles-induced apoptosis by activating TRAIL (tumor necrosis factor (TNF)-related apoptosis-inducing ligand) pathway. In addition, it was reported that these nanovesicles reduced tumor growth in CML (chronic myelogenous leukemia) xenograft mice model (Raimondo et al. 2015). In another study, garlic-derived extracellular vesicles (EVs) exhibited anti-proliferative activity against

kidney (A498) and lung (A549) cancer cells while showing no cytotoxicity in normal human dermal fibroblast (HDF) cells. It was observed that treatment with garlic EVs upregulated the expression of pro-apoptotic genes (p53, Bax, Caspase3 and Caspase9) while downregulated the anti-apoptotic genes (Özkan et al. 2021). In a similar study, fingerroot (*Boesenbergia rotunda*)-derived nanovesicles (FDNVs) were observed to be selectively taken up by colorectal cancer cells (HT-29 and HCT116) but not by normal human colon epithelial cells (CCD 841 CoN). FDNVs inhibited colon cancer cell proliferation by inducing apoptosis, which was mediated by the upregulation of Bax, caspase-3, and caspase-9, along with the downregulation of Bcl-2 expression (Wongkaewkhiaw et al. 2022).

In this context, *Acorus calamus*-derived exosome-like nanoparticles (ACENPs) represent a novel subclass of PDENPs with distinct bioactive properties. While traditional extracts and metal nanoparticles derived from *A. calamus* have demonstrated anticancer effects (Peng et al. 2024), including apoptosis induction and cell cycle arrest in various cancer models, the potential of its exosome-like nanoparticles remains underexplored. Given the unique phytochemical composition of *A. calamus*, ACENPs may offer enhanced cellular uptake, selective internalization into cancer cells, and potent apoptosis-inducing effects similar to or potentially superior to other PDENPs.

Acorus calamus Linn., commonly known as vacha, belongs to the *Acoraceae* family and is a native of North America and northern and eastern Asia. It has been used to treat various diseases, including neurological, respiratory, metabolic, and liver disorders. Rhizome of *A. calamus* has been used to treat cancer in ancient Tibetan and Chinese systems of medicine (Burang 1979; Sharma et al. 2020).

Several recent reports have discussed the anticancer potential of *A. calamus* L. (Haghighi et al. 2017; Thakkar et al. 2022, 2023). A group of researchers reported that an alcoholic extract of *A. calamus* inhibited gastric cancer (AGS) cell proliferation and induced G1-phase cell cycle arrest. In addition, the extract demonstrated anti-angiogenic potential, as evidenced by a reduced tube-like structure formation by HUVEC cells (Haghighi et al. 2016). In another study, a hydromethanolic extract of *A. calamus* inhibited the proliferation of lung cancer (A549) cells by modulating the mitochondrial membrane potential and triggering apoptosis via TRAIL-mediated pathway (Thakkar et al. 2022). Continuing with this study, the research group reported that α -asarone is the main bioactive component involved in inhibiting lung cancer (A549) cell growth (Thakkar et al. 2023). A separate study showed that β -asarone suppresses migration, invasion, and EMT in bladder cancer cells (253 J and 5637) through activation of the ATF6 branch of the ER stress pathway (B. Liu et al. 2023).



Building on the anticancer potential of *A. calamus* and its bioactive compounds, researchers have also explored its nanoparticle-based applications. In a study, gold nanoparticles were synthesized using *A. calamus* extract (AC-AuNPs). These AC-AuNPs exhibited a dose-dependent inhibition of K562 leukemia cells (Jannathul Firdhouse et al. 2024). Another study reported that silver nanoparticles synthesized using *A. calamus* rhizome extract exhibited a dose-dependent inhibition of epidermoid carcinoma cells (A431) (Nayak et al. 2015). A similar study on *A. calamus*-derived silver nanoparticles observed apoptosis in A549 lung carcinoma cells, as evidenced by a decrease in mitochondrial membrane potential, an increase in mRNA levels of Bax and caspases-3, -8, and -9, and a decrease in mRNA levels of Bcl-2 (Thakkar et al. 2024). Another study reported that reduced graphene oxide nanoparticles synthesized using *A. calamus* extract exhibited concentration-dependent inhibition of MCF-7 breast cancer cells. (Smina et al. 2021).

Previous studies have established the antitumor properties of *A. calamus* extracts and purified compounds across various cancer cell lines. However, the impact of *A. calamus*-derived exosome-like nanoparticles (ACENPs) on breast cancer remains unexplored. Therefore, this study aims to evaluate the anticancer properties of ACENPs and investigate their potential as a novel biotherapeutic for breast cancer. We isolated and characterized ACENPs, followed by an assessment of their anti-proliferative activity against ER/PR-positive (MCF-7), HER2-positive (MDA-MB-453), and triple-negative (MDA-MB-231) breast cancer cells.

Materials and methods

Cell culture

MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells were cultured in DMEM (Dulbecco's modified Eagle media) (Gibco, USA) supplemented with 10% fetal bovine serum (Gibco, USA) and 1% pen strep antibiotic (HiMedia). All cells were incubated at 37 °C in a humidified atmosphere with 5% CO₂.

Plant material

500 g of fresh roots of *Acorus calamus* were procured from a local nursery. The root of this plant was identified by an Ayurvedic practitioner.

Isolation of ACENPs

200 g of fresh roots were washed with distilled water to remove dust and soil. The juice of the root was prepared with the help of a blender and filtered through a sieve to remove

large debris. Clarified juice was subjected to sequential centrifugation (8000 g and 20,000 g for 60 min) to remove debris. The supernatant was collected, and ultracentrifugation (Beckman Coulter, Optima XPN-100, USA) was carried out at 120,000 × g for 2 h. The supernatant was discarded, and the obtained pellet containing extracellular vesicles was dissolved in PBS and sterilized by filtration (You et al. 2021). The concentration of nanoparticles was obtained through NTA analysis. All experiments were conducted using a single batch of ACENPs to ensure consistency and reproducibility.

Characterization of ACENPs by TEM and NTA

Nanoparticle size distribution and concentration were analyzed using the NanoSight NS3000 (Malvern, U.K.) equipped with a sCMOS camera. The sample was diluted 1000-fold in PBS buffer before measurement to ensure an optimal particle concentration within the instrument's linear range for accurate tracking. Prior to analysis, the instrument was calibrated using 100 nm polystyrene beads. Measurements were conducted at a controlled temperature of 25.4–25.6 °C. The camera level was set between 14 and 15, with a shutter speed of 1206–1259 and a gain of 245. Data acquisition was performed at 25 frames per second, capturing 1498 frames per measurement. The detection threshold was set to 5 for analysis. Each sample was measured in triplicate, with at least 3000 valid tracks per run to ensure reproducibility. The nanoparticle concentration was reported in particles/mL, and size distribution data were presented as mean and standard deviation. All data were processed using NanoSight NTA 3.4 software, with automatic blur and minimum track length filters applied to minimize background noise and enhance measurement accuracy.

For TEM, carbon-coated copper grids were incubated with ACENPs at room temperature for 5 min and then washed thrice with PBS. Grids were negatively stained with 2% uranyl acetate for 5 min. Samples were dried at room temperature and visualized using a JEM-1400 electron microscope (JEOL, Japan). At least 25 individual particles were measured to ensure statistical robustness, and the mean particle diameter was reported.

Intracellular uptake assay

To monitor the intracellular uptake of plant-derived nanoparticles by human breast cancer cells, nanoparticles were stained with Nile Red fluorescent dye. Breast cancer cells were seeded at a density of 0.1×10^6 cells per well in a 12-well plate. Stained nanoparticles were incubated with breast cancer cells for 24 h. Following treatment, media was aspirated, and cells were washed with PBS to remove unbound particles and then fixed with 4% paraformaldehyde.

Cells were washed with PBS, counterstained with DAPI and visualized using a fluorescence microscope. Images were captured from six different view fields. The experiment was repeated three times.

Anti-proliferative activity

The anti-proliferative activity of ACENPs on breast cancer cells was evaluated using an MTT assay. Breast cancer cells were seeded in 96-well plates at a density of 10^4 cells/well. Cells were treated with different concentrations of nanoparticles for 48 h. Post-treatment media was aspirated, and cells were washed with PBS. Cells were incubated with media containing MTT (0.5 mg/mL) for 4 h at 37 °C. Formazan crystals were dissolved in DMSO, and absorbance was measured at 590 nm using a plate reader (BioTek Cytation 5) equipped with Gen5 software. The experiment was performed in triplicate, and the result was expressed as mean $\pm$ SEM.

Acridine orange–ethidium bromide (AO/EB) staining

Apoptosis was qualitatively assessed following treatment with ACENPs using acridine orange–ethidium bromide dual fluorescent staining. After treatment, breast cancer cells (0.05×10^6 cells per well) were stained with AO/EB dual fluorescent stain for 15 min. Cells were washed with PBS to remove excess stains. Cells were observed using a fluorescent microscope at 100 $\times$ magnification. Images were captured from three different camera positions. AO/EB staining experiment was repeated thrice.

Nuclear changes using DAPI staining

Breast cancer cells were seeded in 12-well plates at a density of 0.05×10^6 cells per well. After 24 h, the cells were treated with various concentrations of nanoparticles for 48 h. Following treatment, the cells were washed with PBS and stained with DAPI (1 μ g/ml) for 15 min. After staining, the cells were washed again with PBS to remove excess stain and examined under a fluorescence microscope. Images were captured from five different camera positions, and the number of apoptotic nuclei was counted using ImageJ. The experiment was replicated three times.

Quantitative analysis of apoptosis using Annexin V-FITC/PI staining

Quantitative analysis of apoptosis induced by ACENPs was performed using annexin V-FITC/PI staining. Breast cancer cells were seeded in 6-well plate at a density of 0.3×10^6 cells per well. Following a 48-h treatment, the cells were trypsinized and washed with PBS. The cell pellet was

resuspended in 100 μ L of annexin V binding buffer and incubated at room temperature for 10 min. The cells were then stained with annexin V-FITC for 10 min in the dark, followed by the addition of PI and further incubation in the dark for 30 min. After incubation, 10,000 cells were analyzed using a flow cytometer (BD Fortessa X20). The experiment was performed in triplicates.

Western blot analysis

After nanoparticle treatment, breast cancer cells were lysed in RIPA (radio-immunoprecipitation assay) buffer. Protein concentration was measured with BCA (bicinchoninic acid protein assay). Proteins were separated by SDS-PAGE and transferred onto the PVDF membrane. Membranes were incubated with primary antibodies (Bax, Bcl-2, GAPDH) overnight at 4 °C. After washing with TBST, the blots were incubated with horseradish peroxidase (HRP)-conjugated goat antirabbit IgG secondary antibody and kept at RT for 2 h. Post-incubation blots were washed with TBST. The chemiluminescent signal was detected using an ImageQuant LAS 500. Densitometric analysis was done with ImageJ software (version 1.53), and protein expression levels were normalized to GAPDH under identical conditions. The experiment was repeated three times.

Metabolic profiling

The metabolic content of ACENPs was analyzed using HR-LC/MS–MS at the Indian Institute of Technology Bombay (IIT Bombay), India. The chromatographic separation and LC–MS analysis were conducted following established protocols with some modifications (Méndez-López et al. 2020).

Statistical analysis

All data were analyzed using GraphPad Prism 7.0 (GraphPad, USA). Results are presented as mean $\pm$ standard error of the mean (SEM) from three independent experiments ($n=3$). Statistical significance was determined using one-way and two-way analysis of variance (ANOVA), followed by Tukey's post hoc test for multiple comparisons. A p value of <0.05 was considered statistically significant. For cytotoxicity assays, the IC₅₀ values were calculated using nonlinear regression analysis. DAPI fluorescence intensity was quantified using ImageJ software (NIH, USA). Western blot densitometric analysis was conducted using ImageJ, with protein expression normalized to GAPDH as a loading control.

Results

Isolation and characterization of exosome-like nanoparticles from *Acorus calamus* using transmission electron microscopy (TEM) and nanoparticle tracking analysis (NTA)

To evaluate the efficiency of ultracentrifugation in isolating *Acorus calamus*-derived exosome-like nanoparticles, we started with 200 g of fresh root material, which was homogenized using a juicer. Following differential centrifugation steps, the final ultracentrifugation at $120,000 \times g$ for 2 h resulted in a pellet weighing 100 mg. The weight-based recovery yield was 0.05%, indicating that only a small fraction of the initial biomass was retained as nanosized vesicles.

The size distribution and concentration of *Acorus calamus*-derived exosome-like nanoparticles (ACENPs) were analyzed using nanoparticle tracking analysis (NTA). The mean hydrodynamic diameter of the nanoparticles was found to be 122.4 ± 5.0 nm, with a mode size of 80.8 ± 3.9 nm, indicating that the most frequently occurring nanoparticle size is approximately 80 nm. To further assess the size distribution, percentile values were determined. The D10, D50, and D90 values were 74.2 ± 2.9 nm, 106.4 ± 6.1 nm, and 192.8 ± 7.3 nm, respectively. These values indicate that 10% of the particles were below 74.2 nm, 50% were below 106.4 nm, and 90% were below 192.8 nm, suggesting a broad size distribution with a slightly right-skewed pattern. The standard deviation (SD) of 51.4 ± 1.6 nm further confirms the presence of some polydispersity within the sample. The total particle concentration was determined to be $1.58 \times 10^{11} \pm 1.54 \times 10^7$ particles/mL, with an average of 22.7 ± 2.8 particles per frame. The particle center count per frame was 25.0 ± 3.4 , confirming the presence of a well-dispersed population of nanoparticles. These results suggest that ACENPs exhibit a size distribution consistent with exosome-like nanoparticles, with a dominant fraction in the 80–122 nm range. The broad distribution, with some larger particles extending towards 192.8 nm, indicates the presence of heterogeneous vesicle populations, which may arise from the natural biogenesis of plant-derived extracellular vesicles. (Fig. 1a). Next, we performed transmission electron microscopy (TEM) of nanoparticles to assess their morphology. Isolated nanoparticles showed spherical morphology with an average diameter of 134.6 ± 5.9 nm (Fig. 1b). The results indicate ultracentrifugation effectively concentrated exosome-like nanoparticles from the crude extract while removing larger cellular debris and unwanted macromolecules. The relatively low yield is expected due to the selective enrichment of nanosized vesicles over bulk plant material. Future studies optimizing centrifugation parameters and purification steps could further improve recovery efficiency.

Next, we lysed the ACENPs in RIPA buffer and quantified protein using BCA assay. The protein concentration was 0.7213 ± 0.0088 $\mu\text{g/mL}$. We determined the purity of isolated ACENPs as reported previously (Webber & Clayton 2013). The obtained nanoparticles showed excellent purity (6.44×10^{11} particles/ μg) determined by the particle-to-protein concentration ratio.

To deliver their molecular cargo inside the cell, nanoparticles must first interact with the cell membrane. Therefore, we assessed the internalization ability of isolated ACENPs in breast cancer cells. To confirm nanoparticle uptake, breast cancer cells were incubated with Nile Red-stained ACENPs at a concentration of 2×10^{11} particles/mL for 24 h. Following treatment, the cells were washed and counterstained with DAPI. Fluorescence microscopy confirmed that ACENPs successfully penetrated the cytoplasm of breast cancer cells (Fig. 1c).

Cytotoxicity of ACENPs on breast cancer cells

To evaluate the cytotoxicity of *Acorus calamus*-derived exosome-like nanoparticles (ACENPs), MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells were treated with varying concentrations of nanoparticles, ranging from 1×10^7 to 2×10^{10} particles/mL. At low concentrations, ACENPs did not exhibit significant cytotoxicity in MCF-7 and MDA-MB-231 cells. However, at higher concentrations, a significant reduction in cell viability was observed in both MCF-7 and MDA-MB-231 cells (Fig. 2a, c). At the highest concentration, cell viability decreased by 17.6% in MCF-7 cells and 25% in MDA-MB-231 cells. In contrast, MDA-MB-453 breast cancer cells exhibited significant cytotoxicity at all tested concentrations (Fig. 2b), with a 35.8% reduction in cell viability at the highest concentration.

Acorus-derived nanoparticles (ACENPs)-induced apoptosis in cancer cells

Cancer cells resist apoptosis, which enables them to contribute to carcinogenesis and drug resistance. MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells were treated with ACENPs for 48 h to assess their apoptosis-inducing potential using various methods.

Following treatment with nanoparticles at different concentrations, cells were stained with Acridine Orange–Ethidium Bromide (AO/EB) dual fluorescent stain and observed under a fluorescence microscope. Viable cells appeared green, as acridine orange penetrated the intact membrane of these cells. In contrast, apoptotic cells appeared yellow–orange, as ethidium bromide can only enter through the damaged membrane of apoptotic cells. Almost all the cells in the control group appeared green, while cells treated with ACENPs displayed yellow–orange fluorescence. A

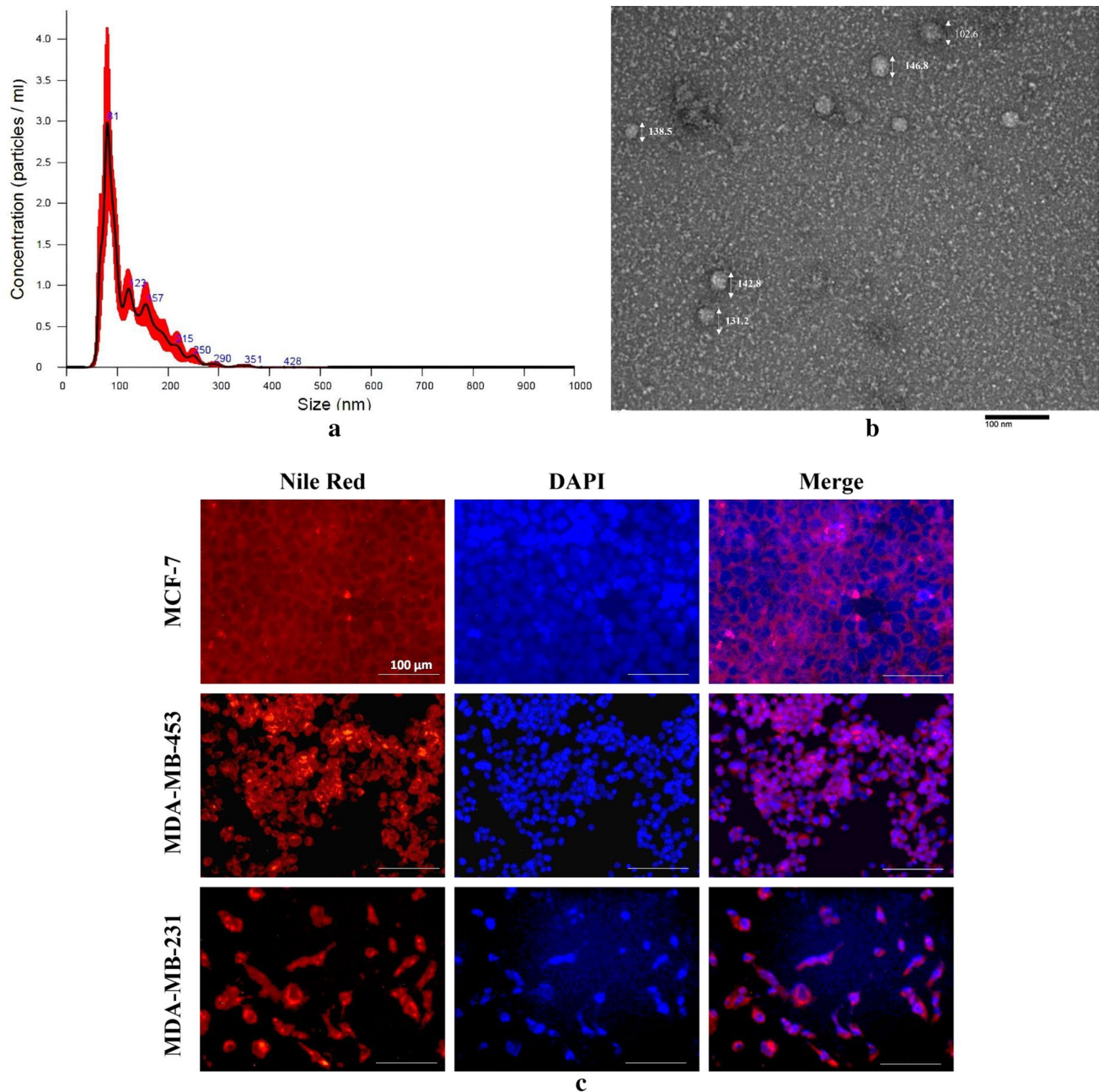


Fig. 1 Characterization of *Acorus calamus*-derived exosome-like nanoparticles (ACENPs) and their cellular uptake. **a** Nanoparticle tracking analysis (NTA) was used to determine the size distribution and concentration of ACENPs. **b** Transmission electron microscopy (TEM) was employed to assess the morphology of the isolated

ACENPs. **c** ACENPs were stained with Nile Red fluorescent dye and incubated with MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells. Fluorescent images were captured using a fluorescence microscope, where red indicates ACENPs and blue represents nuclei. Data are presented as mean $\pm$ SEM ($n = 3$)

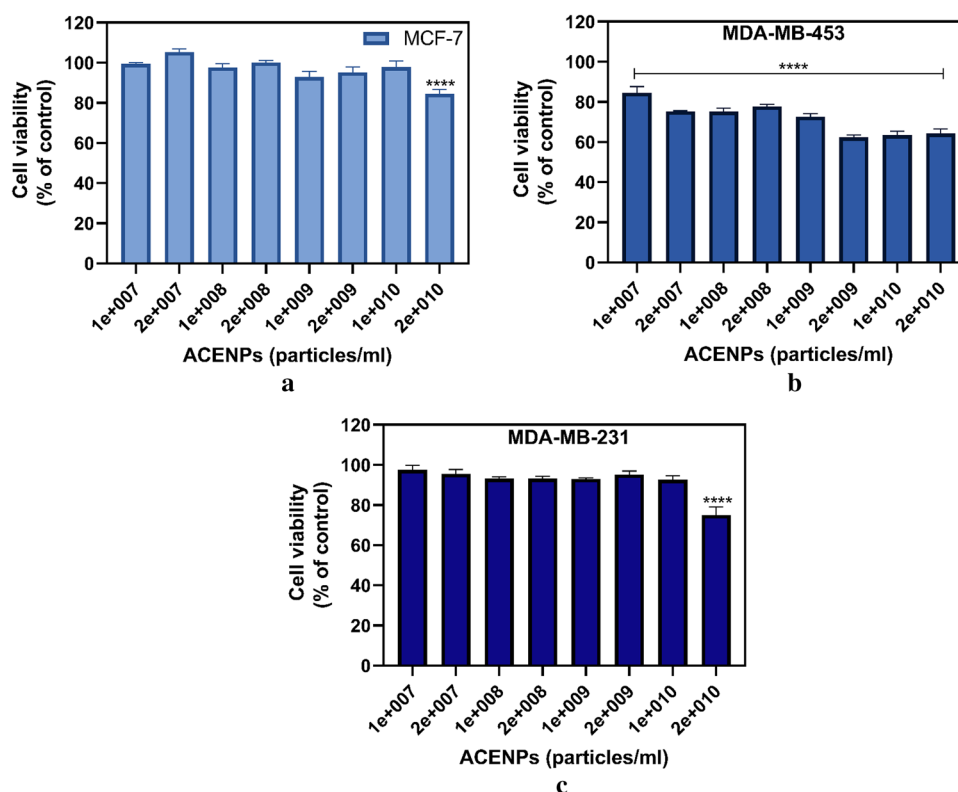
concentration-dependent apoptotic effect was observed in all treated cells (Fig. 3a–c).

In addition to AO/EB staining, we performed DAPI staining to observe the nuclear changes occurring during apoptosis. Cells were treated with varying concentrations of nanoparticles for 48 h and stained with DAPI. After staining, cells were visualized under a fluorescence microscope.

Fragmented and deformed nuclei were observed in the treatment group, while normal nuclear morphology was observed in the untreated control group (Fig. 3d, e).

Furthermore, we examined the changes in the expression of apoptosis-related genes, namely Bax and Bcl-2. Following treatment with ACENPs, an increase in the expression of Bax (pro-apoptotic protein) was observed. In contrast, Bcl-2

Fig. 2 Anti-proliferative activity of ACENPs in breast cancer cells. **a** MCF-7, **b** MDA-MB-453, and **c** MDA-MB-231 breast cancer cells were treated with ACENPs at varying concentrations, and cytotoxicity was assessed using the MTT assay after 48 h. Experiments were performed in triplicates ($n = 3$). Statistical significance between the control and treatment groups was determined using one-way ANOVA (**** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, ns: non-significant). Results are expressed as mean $\pm$ SEM



(anti-apoptotic protein) was downregulated in a concentration-dependent manner (Fig. 3f–h). A study reported that Bax/Bcl-2 ratio indicates the cell's propensity to promote or resist apoptosis (Raisova et al. 2001). In our study, treatment with ACENPs significantly increased the Bax/Bcl-2 ratio in all treated cells compared to the untreated control (Fig. 3f–h).

To further confirm the apoptosis induction by ACENPs, breast cancer cells were stained with Annexin V-FITC and propidium iodide (PI) and analyzed by flow cytometry to quantitatively assess the percentage of live, necrotic, and apoptotic cells. Following treatment with ACENPs, the percentage of apoptotic cells in MCF-7 cells increased from 3.2% to 5.7% with increasing nanovesicle concentration. A similar trend was observed in MDA-MB-453 and MDA-MB-231 cells. Notably, MDA-MB-453 cells exhibited the highest apoptosis rate, with 40% of cells in early apoptosis and 24.9% in late apoptosis following treatment with ACENPs at the maximum concentration (Fig. 3i).

Metabolomic profiling of ACENPs revealed the presence of several bioactive constituents

Identifying the phytochemicals in ACENPs is essential for determining the chemical constituents responsible for their anticancer effects on breast cancer cells. To analyze the

metabolomic profile of these nanoparticles, we employed liquid chromatography–mass spectrometry (LC–MS). The chromatogram of ACENPs in both positive and negative ionization modes is presented in Fig. 4a.

The metabolomic profiling of ACENPs revealed a diverse range of bioactive compounds (255 compounds in positive ion mode and 85 compounds in negative ion mode), each contributing to distinct therapeutic properties. Table 1 summarizes these metabolites along with their potential biological functions, highlighting their role in antioxidant, anti-inflammatory, and neuroprotective activities. Notable compounds detected in positive ion mode included crotonamide, trigonelline, pipecolic acid, arecoline, sedanolide, venlafaxine, purpurumycin, etc. Meanwhile, the negative ion mode revealed the presence of azelaic acid, distomadine A, asarone, (+)-[6]-gingerol, thapsic acid, queen bee acid, and geranylacetone, among others.

Based on the available literature, the identified phytochemicals are classified into flavonoids, terpenes, alkaloids, and organic acids (Table 2). The structures of select phytochemicals identified in ACENPs were retrieved from the PubChem database (<https://pubchem.ncbi.nlm.nih.gov/>) and are shown in Fig. 4b. The complete list of identified compounds in ACENPs is provided in Supplementary Data Files 1 and 2.

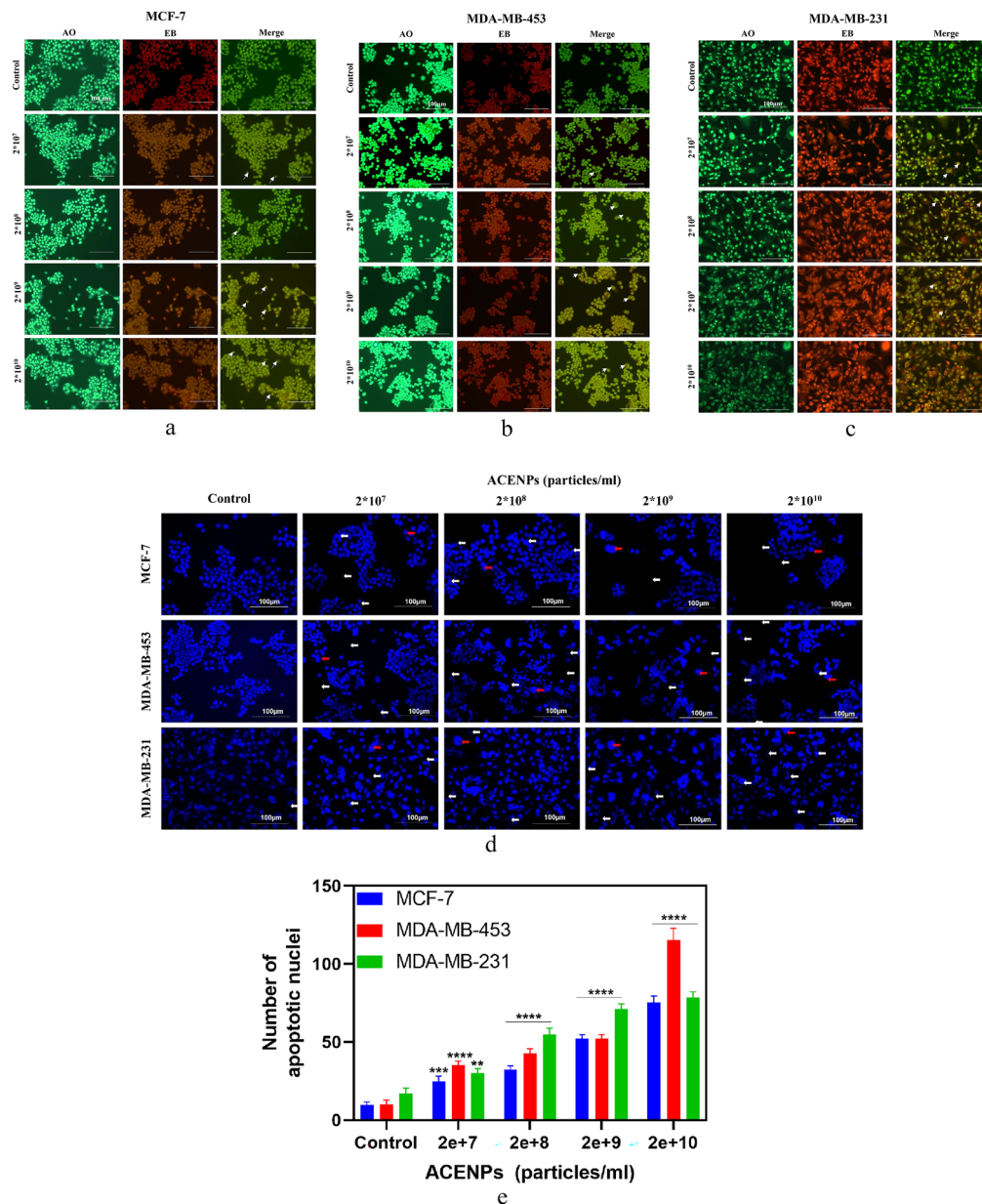


Fig. 3 *A. calamus*-derived nanoparticles (ACENPs)-induced apoptosis in breast cancer cells. Apoptosis-associated morphological changes were observed in cancer cells following treatment with varying concentrations of ACENPs. Representative images from AO/EB dual fluorescent staining illustrate apoptotic cells (white arrows) in **a** MCF-7, **b** MDA-MB-453, and **c** MDA-MB-231 cells (scale bar = 100 μ m). **d** DAPI staining highlights nuclear alterations in cancer cells after ACENPs treatment. White arrows indicate apoptotic nuclear changes, while red arrows highlight necrotic cells. Scale bar = 100 μ m. **e** Quantification of apoptotic nuclei, presented as mean $\pm$ SEM, based on counts from five fields (200–300 nuclei per field) in a representative experiment using DAPI staining. Western blot analysis dem-

onstrates the expression patterns of apoptotic markers Bax and Bcl-2 in **f** MCF-7, **g** MDA-MB-453, and **h** MDA-MB-231 cells following nanoparticle treatment. The Bax/Bcl-2 ratio was quantified using ImageJ software and is presented as bar graphs corresponding to panels (**f**), (**g**), and (**h**). **i** The Annexin V-FITC/PI dual-staining assay was performed to assess the percentage of live, apoptotic, and necrotic cells using flow cytometry. The images presented are representative of three independent experiments. Data are expressed as mean $\pm$ SEM ($n = 3$). Statistical significance was determined using one-way ANOVA with multiple comparisons (**** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$)

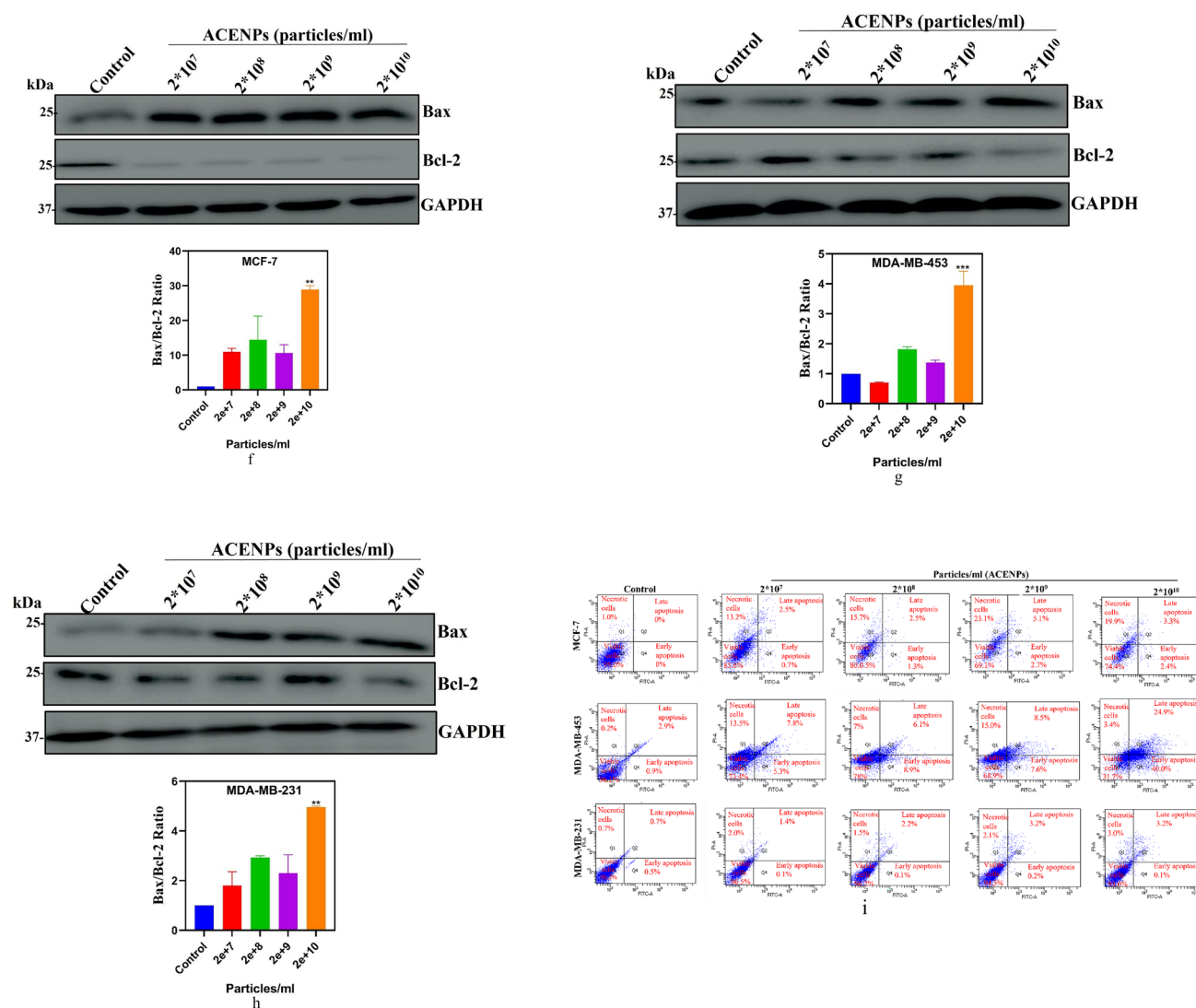


Fig. 3 (continued)

Discussion

The rising drug resistance to existing therapies underscores the need for novel treatment strategies for breast cancer. Although various research studies are available on the anticancer potential of plant extracts (Deng et al. 2017; Gansukh et al. 2019; González-González et al. 2023; Haghighi et al. 2017; Seetharamaiah et al. 2023), their complex phytochemical profile and non-selectivity towards cancer cells are major issues. To address this problem, multiple researchers reported the anticancer properties of plant-derived exosome-like nanoparticles (Özkan et al. 2021; Raimondo et al. 2015; Wongkaewkhiaw et al. 2022). These exosome-like nanoparticles contain bioactive phytochemicals and proteins enclosed by a phospholipid bilayer (Suharta et al. 2021; Zhou et al. 2023). Upon interacting with recipient cells, these exosome-like nanoparticles deliver their molecular cargo

and exert physiological effects. Various reports are available regarding their ability to selectively target cancer cells (Özkan et al. 2021; Raimondo et al. 2015).

Acorus calamus Linn. belongs to the *Acoraceae* family and is used in the Ayurvedic system to treat depression, anxiety, Alzheimer's, and cancer (Haghighi et al. 2017; Peng et al. 2024; Rai et al. 2023; Thakkar et al. 2022). While a few reports have documented the anticancer properties of *A. calamus* extract, no studies have explored the anticancer potential of *A. calamus*-derived exosome-like nanoparticles (Haghighi et al., 2017; Thakkar et al. 2022). Thus, in the present study, we reported the anticancer activity of *A. calamus*-derived exosome-like nanoparticles (ACENPs) in breast cancer cells.

Although various techniques are available to isolate exosomes, we used ultracentrifugation to isolate the nanoparticles from *A. calamus* to obtain pure nanoparticles

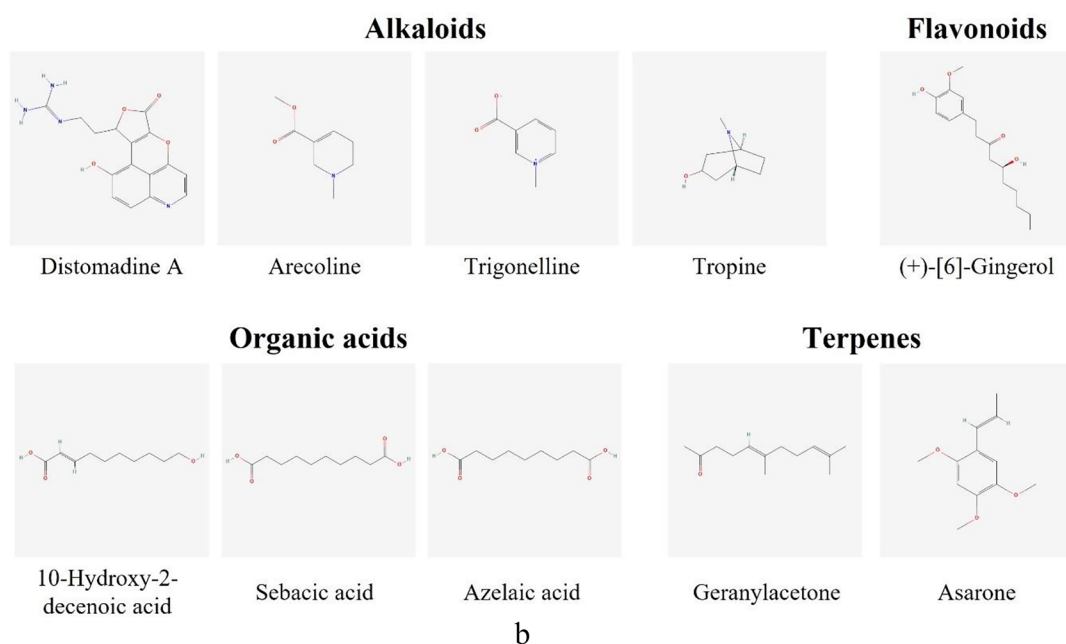
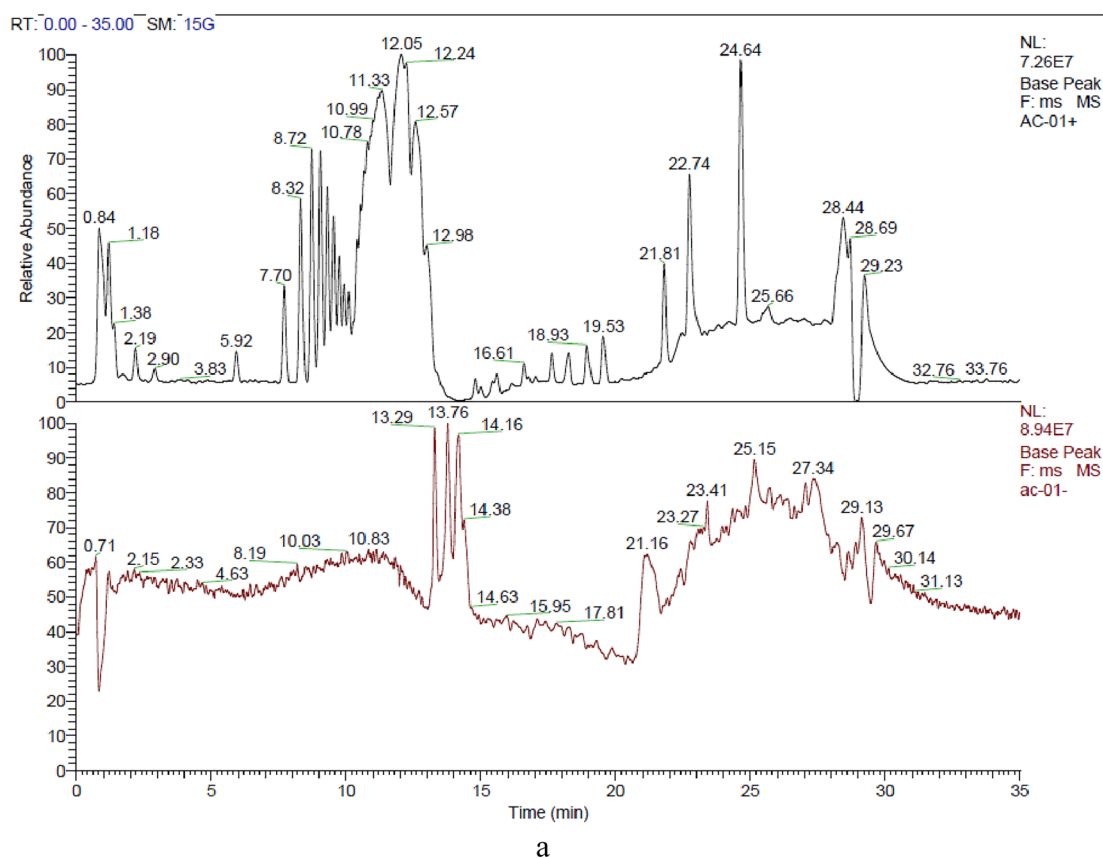


Fig. 4 LC–MS analysis of *Acorus calamus*-derived exosome-like nanoparticles (ACENPs). **a** Total ion chromatogram of ACENPs obtained using liquid chromatography–mass spectrometry (LC–MS)

in both positive and negative ionization modes, showing the detected compounds. **b** Chemical structures of selected bioactive compounds identified in ACENPs through LC–MS analysis

Table 1 Summary of metabolomic profiling of ACENPs and their potential biological functions

Metabolite	Class/category	Potential biological function	Reference(s)
Phenethylamine	Alkaloid	Anticancer, antimicrobial, antioxidant	(Cloonan et al. 2009; Yildirim et al. 2022)
Betaine	Amino acid	Anti-inflammatory	(Chen et al. 2022)
Nootkatone	Sesquiterpene ketone	Anticancer	(Yoo et al. 2020)
Trigonelline	Alkaloid	Anti-diabetic	(YOSHINARI et al. 2009)
Arecoline	Alkaloid	Endocrine function modulator	(Dasgupta et al. 2020; Volgin et al. 2019)
Monolinolenin	Fatty acid	Antibacterial	(Kusumah et al. 2020)
Pipecolic acid	Carboxylic acid derivative of piperidine	Plant immunity	(Brambilla et al. 2023; Cai et al. 2024)
Sedanolid	Tetrahydropthalimide	Anti-oxidative, anti-inflammatory	(Tabei et al. 2023)
Caryophyllene oxide	Sesquiterpenoid oxide	Anticancer	(Shabana et al. 2023; Xiu et al. 2022)
4-Methoxycinnamic acid	Phenolic acid	Antifungal	(Wang et al. 2023)
Azelaic acid	Dicarboxylic acid	Anti-psoriasis	(Li et al. 2022)
Asarone	Terpenes	Anticancer, neuroprotective	(Kalra et al. 2025; Liu et al. 2023; Meng et al. 2021; Thakkar et al. 2023)
Suberic acid	Dicarboxylic acid	Anti-aging,	(Kang et al. 2019, 2022)
Sebacic acid	Dicarboxylic acid	insecticidal, antimalarial, antileishmanial	(Alkhaibari and Alanazi 2022)
(+)-[6]-Gingerol	Flavonoids	Anti-inflammatory, anticancer, antioxidant	(Fan et al. 2015; Ghosh et al. 2023; Pan et al. 2023; Umar et al. 2023)
Traumatic Acid	dicarboxylic acid	Anti-oxidant, anti-lipogenesis	(Gao et al. 2023; Jabłońska-Trypuć et al. 2016)
Geranylacetone	Terpenes	Trypanostatic	(Saad et al. 2019)
Dodecanedioic acid	Dicarboxylic acid	Hypolipidemic	(Angelini et al. 2024)

Table 2 Classification of selected metabolites identified in ACENPs via LC–MS

Phytochemical name	Retention time (min)	Formula	Observed m/z	References
Alkaloids				
Distomadine A	7.662	$C_{16}H_{14}N_4O_4$	326.10051	(Pearce et al. 2003)
Arecoline	0.992	$C_8H_{13}NO_2$	155.0945	(Liu et al. 2016)
Trigonelline	0.922	$C_7H_7NO_2$	137.0475	(Nugrahini et al. 2020)
Tropine	0.961	$C_8H_{15}NO$	141.11527	(Kohnen-Johannsen and Kayser 2019)
Flavonoids				
(+)-[6]-Gingerol	14.745	$C_{17}H_{26}O_4$	294.18352	(Ghosh et al. 2023)
Terpenes				
Geranyl acetone	15.463	$C_{13}H_{22}O$	194.16665	(Bonikowski et al. 2015)
Asarone	14.193	$C_{12}H_{16}O_3$	208.10946	(Bisht et al. 2011)
Organic acids				
Queen bee acid	11.379	$C_{10}H_{18}O_3$	186.12491	(Martínez-Chacón et al. 2023)
Sebacic acid	11.783	$C_{10}H_{18}O_4$	202.11992	(Iaconelli et al. 2010)
Brassylic acid	14.667	$C_{13}H_{24}O_4$	244.16754	(Guo et al. 2007)
Azelaic acid	10.703	$C_9H_{16}O_4$	188.10426	(Pan et al. 2017)

(Li et al. 2017). We characterized isolated ACENPs using TEM and NTA. The concentration of ACENPs was 1.58×10^{11} particles/ml, while the diameter of ACENPs was 134.6 ± 5.9 nm. Our results were consistent with the previously reported literature on plant-derived nanoparticles (Özkan et al. 2021; Wongkaewkhiaw et al. 2022). Isolated ACENPs showed spherical morphology. The particle-to-protein concentration ratio is used to evaluate exosome

purity, with a high ratio indicating highly purified nanovesicles. We isolated ACENPs with a particle-to-protein ratio of 6.44×10^{11} particles/ μ g, aligning with the reported criteria for pure nanovesicles (Webber & Clayton 2013).

Exosomes must interact with recipient cells to deliver their molecular cargo. Several studies have reported that plant-derived exosomes can be internalized by cancer cells. In a study, ginger-derived exosome-like nanoparticles

(GELNs) are reported to be effectively internalized by triple-negative breast cancer cells (Anusha et al. 2023). In a similar study, extracellular vesicles derived from *Perilla* leaves (Perex) were found to be internalized by MDA-MB-231 breast cancer cells (Kim et al. 2023). No studies have reported the ability of *A. calamus*-derived nanoparticles to interact with breast cancer cells. Here, we demonstrate for the first time that *A. calamus*-derived exosome-like nanoparticles (ACENPs) are effectively internalized by MCF-7, MDA-MB-453, and MDA-MB-231 breast cancer cells. The precise mechanism of plant-derived exosome internalization in breast cancer cells remains unclear. However, a group of researchers reported that MDA-MB-231 breast cancer cells exhibited high expression of caveolin-1 on their surface compared to normal cells, and inhibition of caveolin-1 by filipin significantly decreased the uptake of *Perilla*-leaf-derived extracellular vesicles (Perex) by MDA-MB-231 cells (Kim et al. 2023). These findings suggest that ACENPs might enter breast cancer cells via caveolae-mediated endocytosis.

We evaluated the cytotoxicity of ACENPs on breast cancer cells and observed significant cytotoxic effects at the highest concentration (2×10^{10} particles/mL). In addition, ACENPs exhibit lower cytotoxicity compared to *A. calamus* extracts, which could make them promising candidates for use as drug delivery vehicles (Haghighi et al. 2017; Thakkar et al. 2022). This aligns with previous studies on plant-derived exosomes exhibiting anticancer potential. For instance, a study reported that garlic-derived exosomes reduced the viability of U-87 human glioblastoma cells, neuroblastoma cells (SHSY5Y), prostate cancer cells (PC-3), and pancreatic cancer cells (Panc-1a) by 25–40% at the highest tested concentration (Ünsal et al. 2024). Similarly, lemon-derived exosomes have been shown to inhibit the growth of gastric adenocarcinoma cells (SGC-7901, AGS, and BGC-823) in a dose-dependent manner.

In addition, metal nanoparticle studies have reported varying cytotoxicity levels (Bai Aswathanarayan et al. 2018; Gholami-Shabani et al. 2022). For instance, a study evaluating the anticancer effects of gold, iron oxide, and zinc oxide nanoparticles on HT29 human adenocarcinoma cells found that only zinc oxide nanoparticles effectively inhibited cancer cell growth (Bai Aswathanarayan et al. 2018). Similarly, another study reported that gold nanoparticles synthesized using *Aspergillus kambarensis* failed to inhibit the growth of HepG2 cells (Gholami-Shabani et al. 2022). ACENPs offer a naturally derived alternative with potential therapeutic advantages. While metal nanoparticles often require surface modifications to enhance biocompatibility and stability (Guerrini et al. 2018), ACENPs may provide an intrinsic bioactive framework that supports drug delivery and therapeutic applications. Although their exact mechanisms of action, including immunogenicity and antioxidant

or anti-inflammatory effects, require further investigation, their plant-derived origin suggests a favorable safety profile. These findings further support the potential of ACENPs as promising anticancer agents.

Beyond their anticancer properties, the neurological effects of ACENPs warrant consideration, given the traditional medicinal use of the *A. calamus* in neurological disorders. *A. calamus* and its bioactive constituents exhibit strong neuroprotective effects, suggesting that ACENPs could also possess neuroprotective properties. In a study, β -Asarone mitigates β -amyloid-induced neurotoxicity via the PI3 K/Akt/Nrf2 axis, reducing oxidative stress and apoptosis (Meng et al. 2021). In another study, *A. calamus* extract improves blood–brain barrier integrity and promotes neurological recovery in traumatic brain injury models (Kalra et al. 2025). It also protects against acrylamide-induced neurotoxicity by enhancing glutathione levels and maintaining dopamine receptor balance (Shukla et al. 2002). These findings indicate that ACENPs may offer dual benefits, anticancer effects, and potential neuroprotection, highlighting their significance in biomedical applications while emphasizing the need for further studies to evaluate their safety and efficacy.

For a drug to be effective, it must successfully trigger apoptosis in cancer cells. To address this, we investigated the apoptosis-inducing potential of ACENPs. The acridine orange–ethidium bromide (AO/EB) dual fluorescent staining technique is cost-effective for studying apoptosis. We observed that at low ACENP concentrations, most cells appeared green, indicating viability. As the concentration increased, a majority of cells exhibited a yellow–orange fluorescence, signifying apoptosis. To further investigate nuclear changes associated with apoptosis following ACENP treatment, we performed DAPI staining. Nuclear fragmentation and condensation were observed in all tested cell lines in a dose-dependent manner. We utilized flow cytometry to quantitatively evaluate the apoptosis-inducing potential of ACENPs and observed that over 64% of MDA-MB-453 cells underwent apoptosis upon treatment.

The Bax/Bcl-2 ratio plays a critical role in determining apoptosis susceptibility as shown previously (Raisova et al. 2001). ACENPs modulate this ratio by upregulating Bax, leading to mitochondrial outer membrane permeabilization and cytochrome c release, activating caspase-dependent apoptosis. Simultaneously, ACENP downregulates Bcl-2, reducing mitochondrial integrity and enhancing apoptotic susceptibility (Bageci et al. 2006). Similar to findings in melanoma, a higher Bax/Bcl-2 ratio sensitizes cells to apoptotic stimuli, while a lower ratio confers resistance. This mechanistic insight suggests ACENP as a potential apoptotic inducer with therapeutic applications, warranting further investigation into its mitochondrial-targeting pathways (Raisova et al. 2001). Similar studies have reported that

plant-derived exosomes can induce apoptosis in cancer cells (Özkan et al. 2021; Raimondo et al. 2015; Wongkaewkhiaw et al. 2022).

We conducted metabolomic profiling to identify the bioactive constituents responsible for the anticancer properties of ACENPs. Our analysis revealed the presence of α -asarone, venlafaxine, 10-hydroxy-2-decenoic acid (Queen bee acid), trigonelline, gingerol, and other metabolites in ACENPs. Previous studies have demonstrated that α -asarone inhibits the proliferation of A549 lung cancer cells by inducing apoptosis (Thakkar et al. 2023). It exhibited cytotoxic effects in lung carcinoma cells (IC_{50} 21.43 μ M). Furthermore, α -asarone-induced apoptosis was evidenced by a reduction in mitochondrial membrane potential, nuclear condensation, upregulation of Bax and caspase (3, 8, and 9) levels, and downregulation of Bcl-2 expression (Thakkar et al. 2023).

Several studies have reported that 6-gingerol exhibits anticancer activity against various cancer cells (Fan et al. 2015; Sp et al. 2021). A study reported that 6-gingerol induced apoptosis in MDA-MB-231 and MCF-7 breast cancer cells by modulating the Bax/Bcl-2 ratio and promoting cytochrome-c release (Sp et al. 2021). Another study demonstrated that 6-gingerol inhibited the proliferation of osteosarcoma cells by inducing apoptosis through the modulation of Bax and Bcl-2 expression (Fan et al. 2015). Another study reported that trigonelline, in combination with cisplatin/etoposide, inhibited the nuclear accumulation of Nrf2 (Nuclear factor erythroid 2-related factor 2) and significantly suppressed lung carcinoma cell proliferation (Fouzder et al. 2021).

10-Hydroxy-2-Decenoic acid, known as queen bee acid (QBA), is derived from honeybee secretion. Previous studies have reported that QBA exhibits anticancer properties (Albalawi et al. 2021; Saad Al Shehri et al. 2023). QBA significantly inhibited the proliferation of hepatocellular carcinoma cells (HepG2) by inducing apoptosis, as evidenced by the upregulation of pro-apoptotic genes (caspase-3, Bax) and the downregulation of anti-apoptotic genes (Shehri et al. 2023). In a similar study on the anticancer effects of QBA, it was observed that QBA, in combination with cyclophosphamide, significantly reduced tumor size and tumor-associated serum markers in mice bearing Ehrlich solid tumors (Albalawi et al. 2021).

Venlafaxine, one of the metabolites identified in ACENPs, is primarily used to treat depression. However, recent studies have also reported its potential anticancer activity (Niu et al. 2023; Zhang et al. 2023). A study reported that venlafaxine exhibits cytotoxicity in MV3 melanoma cells. In addition, venlafaxine induced apoptosis in melanoma cells in a Nur77-dependent pathway (Niu et al. 2023).

Interestingly, some of the identified compounds in ACENPs have shown the potential to enhance the efficacy

of conventional chemotherapy. For instance, trigonelline has been reported to improve the cytotoxic effects of cisplatin/etoposide in lung cancer cells by inhibiting Nrf2 accumulation (Fouzder et al. 2021). Similarly, QBA, in combination with cyclophosphamide, significantly reduced tumor burden in mice models (Albalawi et al. 2021). Such findings suggest that ACENPs may serve as a synergistic agent in combination therapies, potentially lowering the required doses of chemotherapeutic agents and mitigating their associated toxicities. Future research should investigate the combinatory effects of ACENPs with standard chemotherapy regimens to optimize their therapeutic potential.

By linking our metabolomic findings with existing literature on the anticancer properties of the identified metabolites, this study provides a comprehensive perspective on their potential role as cancer therapeutics.

While ACENPs have not yet been studied in clinical settings, their biocompatibility and therapeutic properties suggest potential applications in nanomedicine. Given the known pharmacological effects of *A. calamus* and the emerging interest in plant-derived exosome-like nanoparticles, ACENPs may hold promise in neurodegenerative disorders. In addition, their anti-inflammatory activities could be explored for managing chronic inflammatory conditions.

The immunogenicity of plant-derived exosome-like nanoparticles is a critical factor influencing their clinical applicability. Since ACENPs are naturally derived, they are expected to exhibit low immunogenicity. However, their interaction with the immune system remains unexplored. Understanding the immunogenic profile of ACENPs will be essential for ensuring their safety and optimizing their therapeutic applications.

As naturally derived nanocarriers, ACENPs could enhance drug bioavailability and therapeutic precision while potentially reducing adverse effects. However, further in vivo and clinical studies are required to establish their safety, efficacy, immunogenicity, and translational potential.

This study has certain limitations that warrant further investigation. First, the lack of in vivo data limits our ability to fully assess the therapeutic efficacy, biodistribution, and safety profile of ACENPs in a physiological setting. Future studies should incorporate well-designed animal models to evaluate their pharmacokinetics, toxicity, and therapeutic potential in a more complex biological environment. In addition, the effective in vivo delivery of ACENPs poses significant challenges, including stability in physiological conditions, potential interactions with plasma proteins, and rapid clearance. Strategies such as surface functionalization with targeting ligands and encapsulation within biocompatible carriers could improve bioavailability and targeted delivery to tumor sites. Optimizing the route of administration and dosage regimen will also be critical for maximizing therapeutic efficacy while minimizing off-target effects.

Another important aspect to explore is the potential for ACENPs as drug delivery vehicles. Given their biocompatibility and nanoscale size, ACENPs could be engineered to encapsulate therapeutic agents, enhancing targeted delivery and controlled drug release. Future studies should focus on optimizing drug-loading efficiency, release kinetics, and stability to maximize ACENPs' potential as a multifunctional therapeutic platform.

Challenges related to the large-scale production of ACENPs, including batch-to-batch consistency, reproducibility, and stability, also need to be addressed for potential clinical applications. Standardizing nanoparticle isolation protocols, optimizing purification techniques, and implementing quality control measures such as physicochemical characterization and functional assays can help ensure reproducibility. In addition, advanced storage and formulation approaches could enhance the stability and shelf life of ACENPs.

Finally, regulatory challenges must be considered for the successful clinical translation of ACENPs. The approval process for nanoparticle-based therapeutics requires rigorous evaluation of their safety, efficacy, and biocompatibility by regulatory agencies. Establishing standardized protocols for toxicity assessment, long-term safety studies, and adherence to Good Manufacturing Practice guidelines will be essential.

By systematically addressing these limitations and challenges, future research can pave the way for ACENPs to become a promising therapeutic approach in cancer treatment.

Conclusion

Our study demonstrates that ACENPs exhibit anti-proliferative activity in breast cancer cells and induce apoptosis as demonstrated by elevated Bax/Bcl-2 ratio. Metabolomic profiling revealed the presence of several bioactive compounds with potential anticancer properties. Further research is needed to elucidate the detailed mechanisms of ACENP internalization and their anticancer effects at the molecular level. In addition, optimizing alternative methods for ACENP isolation is crucial for large-scale production, as ultracentrifugation is not feasible for industrial applications. Scalable isolation techniques should be explored to facilitate broader clinical and commercial use. To validate in vitro findings, future studies should focus on in vivo testing to assess therapeutic efficacy and safety in relevant animal models. Exploring the clinical translation of ACENPs could contribute to the development of novel treatment strategies for breast cancer. Furthermore, investigating potential synergistic effects with existing chemotherapeutic agents and assessing ACENP pharmacokinetics will further support their therapeutic potential.

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Declarations

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Ethical approval This study was conducted in compliance with ethical guidelines for research involving biological materials. The use of *Acorus calamus* plant materials and breast cancer cells were approved by the institutional biosafety committee under approval number IITD/IBSC/2020/02/5.

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